

scripts/autonomous-qa/lib-remote.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)
Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.H (no secret echo), §6.J (failures surface), §6.R (config from .env), §6.U (no sudo) **Classification:** project-specific

Overview

Sourceable library implementing **Approach B (run-on-thinker)**: co-locate the ENTIRE autonomous-QA heavy stack on thinker.local so build (gradle) + containerized KVM emulator (podman) + adb + connectedAndroidTest + backend all share thinker's loopback. This is the only topology where gradle's adb reaches the emulator and the on-emulator client reaches the backend without WAN/adb tunnels — the Containers submodule's pkg/emulator/containerized.go drives LOCAL podman only (no SSH path), so the emulator **MUST** be local to whichever host runs the matrix; that host is thinker (/dev/kvm + 16 cores).

Thin glue over ssh + rsync, keyed on the same identity that authenticates today (~/.ssh/id_ed25519, confirmed via ssh -G).

Functions

Function	Purpose
remote_preflight	Prove reachability + key auth + podman + rsync + /dev/kvm before any heavy work is dispatched; non-zero (with explanation) if thinker is unfit
remote_sync_repo	rsync the repo → thinker (excludes build/.git/recordings, INCLUDES the gitignored .env so the Go backend's LAVA_AUTH_* + tracker creds reach thinker); --delete keeps the remote tree a faithful mirror
remote_run "<cmd>"	Run a command on thinker with cwd = the synced repo; stdout/stderr stream straight back

Function	Purpose
remote_fetch <remote-rel-path> <local-dest>	Pull curated evidence back

Configuration (from .env, gitignored — config, not secrets per §6.R)

Variable	Default	Meaning
LAVA_API_GO_REMOTE_HOST	thinker.local	QA host
LAVA_REMOTE_HOST_USER	milosvasic	SSH user
LAVA_REMOTE_HOST_KEY	~/.ssh/ id_ed25519	SSH identity (path only; contents never read/printed)
LAVA_THINKER_QA_DIR	/home/<user>/ lava-qa	Remote repo location

SSH behavior

Shared options: BatchMode=yes (key-only, never prompt), bounded connect (10 s), StrictHostKeyChecking=accept-new, keepalives, and a persistent ControlMaster (ControlPersist=300, sockets under \$ {TMPDIR: -/tmp}/lava-qa-ssh-ctrl) so the many remote_run calls in a matrix reuse one TCP/SSH session.

Security posture

- §6.H: no secret is ever echoed. .env travels over the encrypted SSH channel, never logged. The key path is config; its contents are never read.
- §6.U: no sudo/su anywhere on either side.
- §6.J: set -euo pipefail; failures surface, never swallowed.

Usage

```
source scripts/autonomous-qa/lib-remote.sh
remote_preflight
remote_sync_repo
remote_run "scripts/autonomous-qa/run-matrix.sh --backend goapi --
           subsets rutracker --queries 1080p"
remote_fetch ".lava-ci-evidence/autonomous-qa/$(date +%F)/goapi/
             summary.md" /tmp/summary.md
```

Companion files

- `scripts/autonomous-qa/run-matrix.sh` — the matrix typically dispatched via `remote_run`
- `scripts/distribute-api-remote.sh` (+ its doc) — the sibling thinker `remote-distribute` path